

Name: \_\_\_\_\_

## Geologic Hazards Lab: Monitoring the Yellowstone Caldera

Modified version of "Yellowstone is Active, but Will it Erupt?" by Hoeven Kraft and Rachel Teasdale as hosted by GETSI

### Part 1

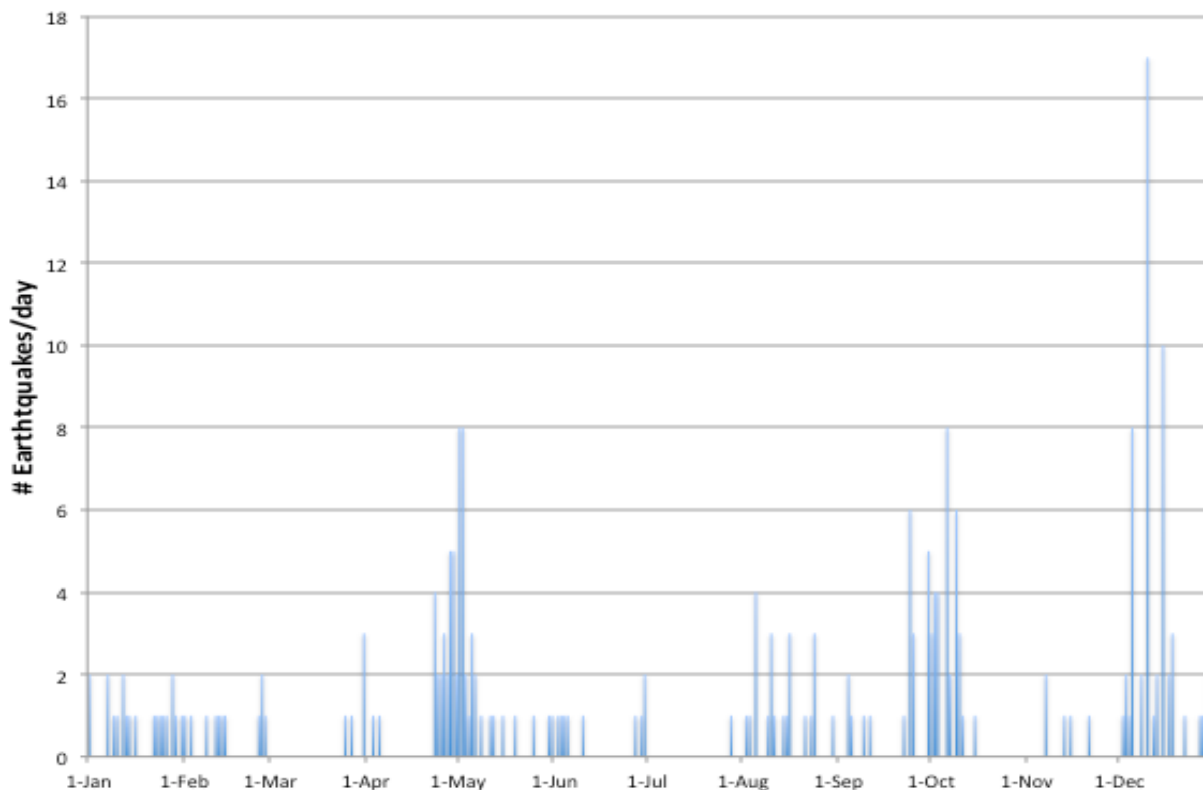
Geologic events happen over very long time spans and occur somewhat randomly. While they are based on systems that occur consistently (e.g., gradual movement of plates results in a release of that stress as an earthquake), there is not a specific time that it should occur or will occur. We try to understand this randomness by determining a mean recurrence interval to represent the average time between events.

Volcanoes are one geologic event for which determining a mean recurrence interval can be helpful to better understand the range of activity, particularly for volcanoes that erupt very infrequently, like **Yellowstone Caldera**. Over the last 2.1 million years, there have been three major eruptions at Yellowstone, resulting in a **Mean Recurrence Interval (MRI) of 600,00-800,000 years**.

Many active volcanoes exhibit **seismic swarms** that may be related to subsurface movement of magma or eruptive activity at the surface. A seismic swarm occurs when there is an **unusually large number of minor earthquakes relative to what is considered to be "background" seismic activity**. Background seismic activity may be occurring regularly in small amounts. A swarm is distinguished by seeing a large number in these small events in a shortened period of time.

Examine the following graph of **earthquakes detected at Mount St. Helens (a stratovolcano) in 2017**.

- As a class, discuss how you would identify seismic swarms in the 2017 data.
- Were these swarms associated with magmatic or volcanic activity in 2017?



## Part 2 Seismic Data at Yellowstone National Park

Your instructor will assign your group a decade of seismic data to examine.

**Which decade did your group receive?**

1990-1999      2000-2009      2010-2018      (circle one)

1) Examine your data. Then **develop a definition of a seismic swarm** using your own words and write it below. For example, consider the size, duration, and indicator that one has started/stopped.

2) Using your working definition above, identify when the seismic swarm(s) occurred during your assigned decade. Record the dates in **Table 1**.

3) **Determine the approximate duration of each seismic swarm** and record the durations in **Table 1**.  
Hint: One increment on the horizontal axis on each graph is 1 month. You can describe the duration as 1/2 month, less than a month, 1 month, 2 months, etc.

4) **Estimate the “size” of each swarm.** In other words, how many earthquakes occurred above the background level of seismic activity. Record each swarm “size” in **Table 1**. Hint: One increment on the vertical axis to the right on each graph (i.e., “cumulative number of earthquakes”) is 200 earthquakes. To simplify things, you can describe the size of the swarm in multiples of 200 using the orange line.

5) Each group should share their results with other groups, allowing them to **complete Table 1**.  
**Calculate the mean recurrence intervals (MRIs)** for the following sizes of seismic swarm.

| Size of Swarm | # of years in record | # of swarms with range | MRI |
|---------------|----------------------|------------------------|-----|
| All sizes     | 40                   |                        |     |
| ≥ 500 EQ      | 40                   |                        |     |
| ≥ 1000 EQ     | 40                   |                        |     |
| ≥ 2000 EQ     | 40                   |                        |     |
| ≥ 3000 EQ     | 40                   |                        |     |

**Table 1**

| Decade         | Date(s) of swarms | Duration of swarm(s): | Size of swarms<br>(How many earthquakes<br>above “background”?) |
|----------------|-------------------|-----------------------|-----------------------------------------------------------------|
| 1980 –<br>1989 | 4/1/1980          | 1 month               | 200                                                             |
|                | 4/1/1981          | ½ month               | 100                                                             |
|                | 3/1/1985          | ½ month               | 200                                                             |
|                | 11/1/1985         | 2 months              | 600                                                             |
|                | 10/1/1989         | 1 month               | 200                                                             |
|                |                   |                       |                                                                 |
|                |                   |                       |                                                                 |
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| 1990 –<br>1999 |                   |                       |                                                                 |
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| 2000 –<br>2009 |                   |                       |                                                                 |
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| 2010 –<br>2018 |                   |                       |                                                                 |
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### Part 3 GPS and InSAR Data at Yellowstone National Park

Your instructor will assign your group a GPS station and its corresponding data (2004-2018) to examine.

**Which GPS station did your group receive?**

NRWY station   OFW2 station   WLWY station   (circle one)

1) Second column of Table 2: Write the **number of seismic swarms and the range of swarm sizes** during the following time intervals. Use the data from Table 1.

- September 2004 – August 2006
- January 2014 – May 2015
- January 2009 – January 2011
- June 2015 – July 2017

2) Third column of Table 2: Describe the **horizontal and vertical movement of the GPS station** during the same four time intervals. **Estimate the amount of total vertical movement.**

3) Fourth column of Table 2: Use InSAR data (i.e., the interferograms) to describe the deformation adjacent to your GPS station. **Use terms like ground inflation, ground deflation, or no change.**

Estimate the **amount of vertical movement** by counting the fringe patterns. Interferograms are available for only two of the time intervals:

September 2004 – August 2006          June 2015 – July 2017

4) Fifth column of Table 2: Write a geological interpretation that combines all data types and address whether the seismic swarms in the four time intervals were associated with the movement of magma.

5) Each group should present their data and interpretations to the rest of the class, allowing everyone to **complete Table 2 for the other GPS stations.**

6) **Are large seismic swarms always related to upward movement of magma?** Use the data in Table 2 to justify your answers.

7) Most Yellowstone earthquakes and seismic swarms are due to groundwater interacting with pre-existing faults within and outside of the Yellowstone Caldera. **What if you knew the precise location of every earthquake swarm and could combine that information with GPS and InSAR data? Could we determine if a swarm is caused by magma or groundwater with more certainty? Explain your reasoning.**

**Table 2**

| <b>GPS Station</b> | <b># of swarms &amp; avg size</b> | <b>GPS Activity</b> | <b>InSAR activity</b> | <b>Geologic Interpretation (swarms due to magma moving up?)</b> |
|--------------------|-----------------------------------|---------------------|-----------------------|-----------------------------------------------------------------|
| <b>NRWY</b>        | Sep. 2004 – Aug. 2006             |                     |                       |                                                                 |
|                    | Jan. 2009 – Jan. 2011             |                     |                       |                                                                 |
|                    | Jan 2014 – May 2015               |                     |                       |                                                                 |
|                    | June 2015 – July 2017             |                     |                       |                                                                 |
| <b>OFW2</b>        | Sep. 2004 – Aug. 2006             |                     |                       |                                                                 |
|                    | Jan. 2009 – Jan. 2011             |                     |                       |                                                                 |
|                    | Jan 2014 – May 2015               |                     |                       |                                                                 |
|                    | June 2015 – July 2017             |                     |                       |                                                                 |
| <b>WLWY</b>        | Sep. 2004 – Aug. 2006             |                     |                       |                                                                 |
|                    | Jan. 2009 – Jan. 2011             |                     |                       |                                                                 |
|                    | Jan 2014 – May 2015               |                     |                       |                                                                 |
|                    | June 2015 – July 2017             |                     |                       |                                                                 |

#### **Part 4 Science Communication**

In the summer of 2021, a large seismic swarm occurred with over 1600 earthquakes (magnitude 3 or less) over the span of a month. Although the earthquakes of the swarm were barely felt, if at all, by people in Yellowstone, the swarm was covered by the media, prompting many people to wonder if we should be worried about the next “supervolcano eruption”.

**How would you respond to a concerned citizen who thought the 2021 seismic swarm was a sign of an impending eruption at Yellowstone?**

In the space below write a response to the hypothetical concerned citizen.

Be sure to include the following points in your response to receive full credit:

- You should include a clear statement about the risk of an eruption.
- Describe the mean recurrence interval (MRI) for swarms of similar size, making sure to explain how to interpret the MRI.
- Explain how GPS motion and InSAR data can be used to detect movement of magma near the surface.
- Qualitatively describe the data and interpretations from Table 2.